

PREPARE to Whisper: A Pretrained Encoder-Based Breakthrough for Accurate, Early and Inclusive AD/ADRD Prediction

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1. Abstract

This report introduces our top-ranking AD/ADRD prediction pipeline, which integrates the OpenAI Whisper pre-trained encoder with a classifier augmented by clinical features. Our solution achieved first place in the modeling stage, demonstrating superior performance compared to methods relying solely on traditional acoustic features or alternative pre-trained models. We detail a robust modeling pipeline that incorporates a two-stage training approach designed to mitigate language bias and enhance generalizability. An ablation study highlights the rationale behind selecting the pre-trained encoder, elucidating the interplay between performance, model parameters, and the pre-training dataset. To enhance model interpretability, we employ a CAM-based explanation method that pinpoints temporal regions critical to decision-making. Finally, we explore future directions, such as the integration of large language models (LLMs) and the use of temporal-level labels, aimed at further advancing the early and inclusive prediction of AD/ADRD.

2. Methodology

2.1 Model Architecture

The overall model architecture is illustrated in **Figure 1**. Our model comprises the following components: **1. Whisper encoder** [1]: Serves as the foundational feature extractor; **2. Projection layer**: Transforms encoder outputs for further processing; **3. Temporal pooling**: Aggregates temporal information from the encoder outputs. Additionally, the **Age** and **Sex** features are integrated into the network by concatenating them with the output of the temporal pooling layer before passing them to the classification layer.

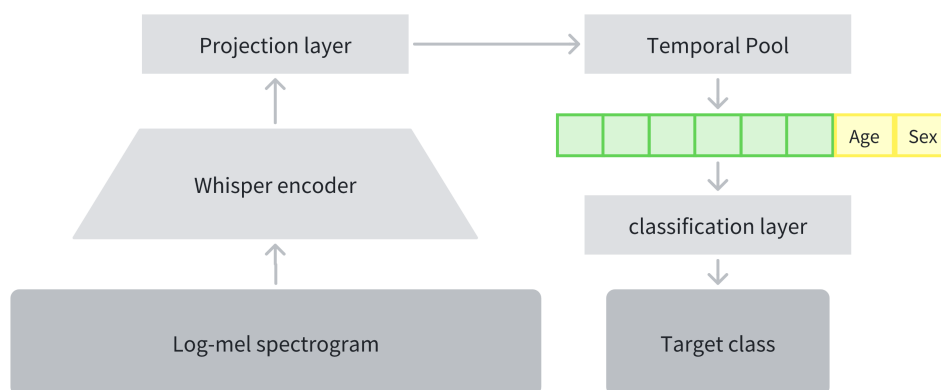


Figure1. Model architecture. The Whisper encoder extracts features with temporal aggregation, and the Age and Sex features are concatenated before the final classification step.

2.2 Model Training process

The training process is structured into two distinct stages to optimize performance and mitigate language bias: **1. Full Training**: The model is initially trained for **2 epochs** using the complete dataset, regardless of language; **2. Language-Specific Training**: The model is fine-tuned separately for **English** and **Spanish** subsets, with language labels inferred using the Whisper ASR model. This two-

stage approach ensures the model captures general patterns across all data while also tailoring performance to language-specific nuances.

To ensure robust evaluation, we employed a stratified K-fold cross-validation approach, where splits were made to balance class distributions across folds. Additionally, we ensured no data leakage by isolating speaker identities between training and validation sets.

2.3 Data augmentation

To enhance model robustness and performance, we employed two data augmentation techniques, implemented using the **audiomentations** package: **1. TimeStretch**: Modifies the speed or duration of the audio signal without altering its pitch, simulating natural variability in speech patterns. **2.**

TimeMask: Randomly silences portions of the audio signal, creating gaps that mimic interruptions or background noise.

These augmentation methods, when combined, contributed to a slight improvement in the model's performance, reducing the log-loss metric by **0.005**.

2.4 Justification for modeling choices

Pretrained acoustic representations have been successfully applied to audio-based AD detection [3, 4]. Self-supervised learned representations, such as **VGGish** and **Wav2Vec 2.0**, consistently outperform traditional acoustic feature sets, including the **ComParE** feature sets [5]. Notably, the **Whisper pretrained model** demonstrates significantly better performance on the ADReSS dataset benchmark. To evaluate the competition dataset, we conducted a benchmark comparing methods based on conventional acoustic features to those utilizing pretrained audio representations. **Table 1** presents these results, highlighting the clear advantage of larger pretraining datasets and more complex models with increased parameters, which lead to superior model performance.

Encoder	Params (Millions)	Pretraining Dataset Size (Hours)	Loss
whisper-tiny	8	680k	0.812
whisper-small	84	680k	0.691
whisper-larger	607	680k	0.630
hubert-large	301	60k	0.701
wav2vec2-base	90	53k	0.721
wavLM	301	94k	0.691
XgBoost	<1	-	0.811

Table 1: Comparison of different modeling method.

2.5 Explainability analysis.

In our temporal model, the input features are processed through an encoder and projection layer to generate hidden states across T time steps. A global average pooling layer aggregates these hidden states into a single pooled representation, which is then passed to a classifier to generate the final

predictions. To interpret the model's decisions, Class Activation Maps (CAMs) are employed, highlighting the contribution of each time step to the final prediction.

CAMs compute the weighted contribution of each time step to the final decision by projecting the hidden states back onto the classifier's weight space. This highlights the temporal regions of the input that are most influential in determining the model's predictions. By visualizing CAMs, we can identify key patterns in the data that the model considers critical, offering greater transparency and interpretability for applications in clinical and research contexts.

3. Performance Analysis

3.1 Overall Model Performance and Metric Choices

Our team achieved **first place** in the modeling stage, with a summary of results presented in **Table 2**:

Team	Log Loss	Rank
NiuBi Team (ours)	0.6299	1
Harris and Kielo	0.6343	2
Salen	0.6523	3

Table 2: Leaderboard Log Loss Ranking

In addition to **log loss**, we evaluated our model using **balanced accuracy** and **F1 score**, which are presented in **Table 3** to provide a comprehensive assessment of model performance.

Table 3: Our best model evaluated with different metrics

	Accuracy	Balanced accuracy	Macro F1
NiuBi Team (ours)	0.72	0.582	0.619

The primary evaluation metric designated by the competition host for the modeling stage was **log loss**. However, recognizing the importance of clinical applicability, we also assessed our model using the **F1 score**, which captures the balance between precision and recall. Furthermore, given the imbalanced nature of the dataset, we employed **balanced accuracy** as an additional metric. Unlike standard accuracy, which can be artificially inflated by predicting the majority class, balanced accuracy accounts for the performance across all classes, providing a more reliable measure in this context.

Our model's robust performance across diverse metrics positions it as a feasible tool for early screening programs in clinical settings. With its ability to handle multilingual data, it can be deployed in community health initiatives, particularly in underserved populations disproportionately affected by AD/ADRD.

3.2 Explainability analysis

We conducted a Class Activation Map (CAM) analysis on our classification model, focusing on an MCI sample (aikh.mp3) with a predicted probability of 0.6109. **Figure 2a** illustrates the log-mel spectrogram of the audio, while **Figure 2b** visualizes the CAM scores, with higher scores indicating regions of greater importance in the model's decision-making process. Additionally, **Figure 2c**

displays the confidence levels derived from the Whisper ASR, providing further context to the analysis.

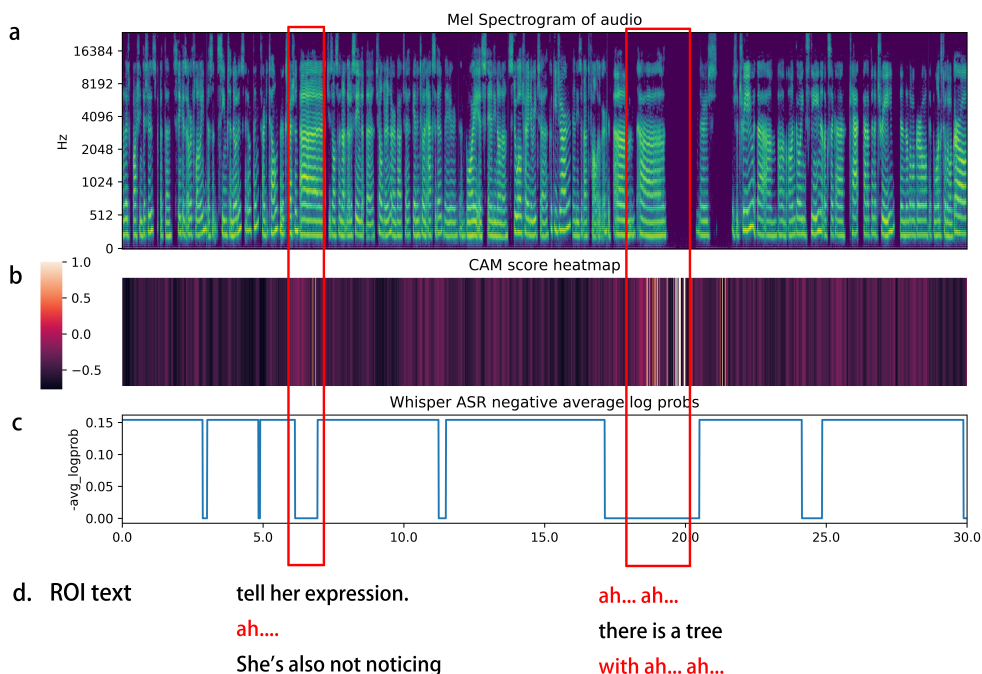


Figure 2. Explainability analysis of the sample audio `aikh.mp3`. The two red regions represent two pronounced pauses, seemingly struggling to recall what to say.

The CAM analysis reveals two notable regions of interest. The first occurs near the **7-second mark**, where a distinct "ah..." expression is observed, aligning with the ASR transcription. The second region spans from **17 to 22 seconds**, during which the speaker demonstrates a clear pause and hesitation, seemingly struggling to recall what to say. This behavior is reflected in two pronounced pauses, with the corresponding transcription text shown in **Figure 2d**.

This analysis underscores the effectiveness of our CAM implementation in uncovering the underlying relationship between speech patterns and clinical prototypes. By identifying these critical regions, our model enhances the transparency and interpretability of its decision-making process, demonstrating its potential for clinical and diagnostic applications.

The CAM analysis not only highlights temporal speech patterns critical to our model's predictions but also aligns with established clinical markers of AD/ADRD, such as prolonged pauses and hesitation in speech. This provides potential diagnostic markers that could enhance early detection efforts in clinical settings.

3.3 Error analysis

The confusion matrix, presented in **Figure 3**, highlights that the majority of errors are due to **false negatives**. This is primarily attributed to the significant imbalance in class labels, where control samples vastly outnumber the MCI and ADRD classes. This imbalance, coupled with the use of the **log loss** metric as the official evaluation criterion—which does not heavily penalize misclassifications of the disease classes—results in the model being more prone to false negatives. A more detailed analysis of these errors and their implications is provided in **Section 4**.

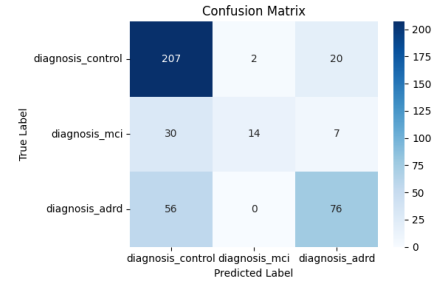


Figure 3. Overall Confusion Matrix.

4. Bias and Ethics Analysis

4.1 Dataset bias analysis

We conducted a decomposition analysis based on the ASR-generated BERT embeddings, as illustrated in **Figure 4a**. The results reveal clear clustering within the dataset, highlighting the presence of **severe batch effects**. Notably, Spanish ($n = 332$) and Mandarin ($n = 43$) samples form distinct clusters separate from English ($n = 1655$) samples. Within the English cluster, additional sub-clusters are evident, reflecting the dataset's diverse origins from **10 different corpora**, which adds to the task's complexity. Of particular concern, all Mandarin audio samples belong exclusively to the MCI class. This introduces a significant bias, as any model trained on this dataset may consistently predict the MCI class for Chinese samples, regardless of their true class.

In addition to these linguistic batch effects, the dataset is imbalanced in several other respects. The healthy cohort ($n = 1140$) is substantially larger than the ADRD ($n = 650$) and MCI ($n = 268$) cohorts (**Figure 4b**). Regarding gender distribution (**Figure 4c**), females ($n = 1219$) outnumber males ($n = 839$). The age distribution is also heavily skewed, with **33.2% of the samples** coming from individuals aged between 80 and 82 (**Figure 4d**).

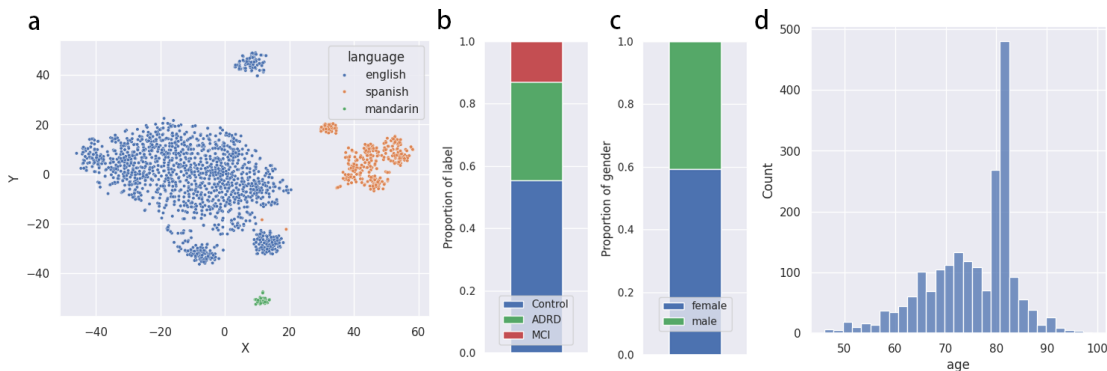


Figure 4. Dataset Bias. (a) Decomposition analysis showing clustering by language and batch effects. (b) Label distribution highlights class imbalance. (c) Gender distribution with more females than males. (d) Age distribution skewed towards individuals aged 80–82.

4.2 Model bias analysis

Our model demonstrated comparable performance for female and male samples, with no statistically significant difference $p = 0.29$ t -test, as shown in **Figure 5a**. Similarly, the Pearson correlation between age and log loss was 0.20, indicating a fairly low correlation (**Figure 5b**). However, the model performed significantly better on English samples ($n = 324$, *test set*) compared to Spanish ($n = 76$) and Galician ($n = 3$), with a statistically significant difference ($p = 4.7 \times 10^{-4}$, t -test) in **Figure 5c**. Notably, all Chinese samples are labeled as MCI, resulting in an artificially low log loss. Consequently, Chinese samples should be excluded from error analysis and practical applications.

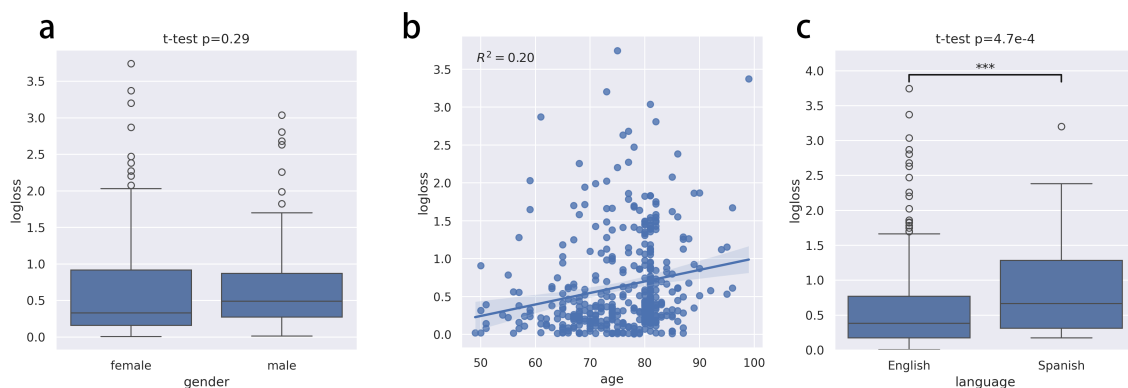


Figure 5: Model Bias Analysis. (a) The model performs similarly across genders with no statistically significant difference. (b) The correlation between age and model performance is low. (c) The model shows a statistically significant performance difference between English and Spanish samples.

Significant performance biases emerged between different corpora (Figure 6), likely due to variations in data collection, linguistic characteristics, and demographics. Differences in audio quality, speaker populations, and recording conditions further contributed to these disparities. Such biases highlight the challenges posed by heterogeneous datasets and underscore the need to mitigate corpus-specific effects, ensuring the model's generalizability and equitable performance across diverse data sources.

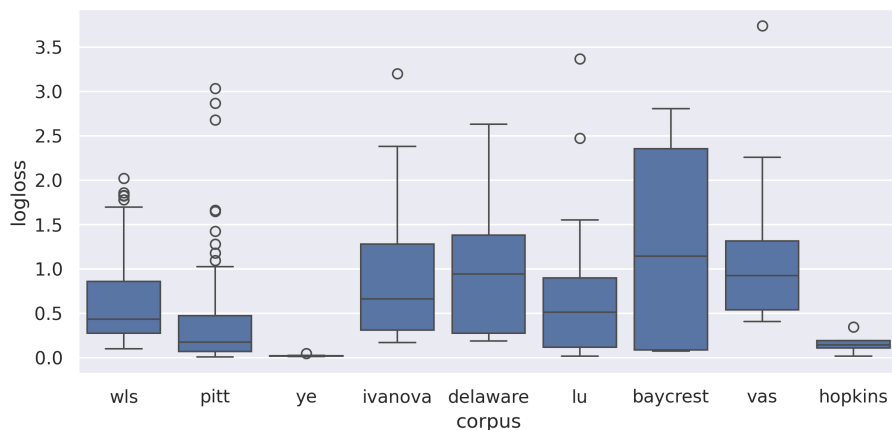


Figure 6: Model Performance Bias Across Different Corpora. Post-competition analysis based on revealed corpus information highlights performance variations between different corpora.

4.3 Subpopulation Error Analysis and Bias Mitigation

Analyzing model performance across languages revealed significant imbalances. The MCI (Mild Cognitive Impairment) class is underrepresented in the English dataset (8%) compared to Spanish (25%), leading to a bias toward predicting Health or ADRD (Alzheimer's Disease and Related Disorders) for English samples. Additionally, all Chinese samples are exclusively MCI, necessitating their exclusion to avoid skewed results.

To mitigate these biases, we employed a two-stage training method (see Section 2.1), which improved log loss and balanced accuracy. Further strategies include:

1. **Balanced Data Collection:** Ensure fair representation across classes and languages.
2. **Metric Adjustment:** Focus on recall for high-sensitivity applications like large-scale audio screening.
3. **Quality Control:** Validate test inputs to meet language and transcription standards.

These measures aim to reduce bias and improve real-world model reliability.

To address the underrepresentation of minority classes, we propose future strategies like targeted data collection to improve class balance and the use of data augmentation to generate synthetic samples for underrepresented subpopulations. These steps will ensure fairer representation and improve generalizability.

5. Future Directions

The main limitations are dataset bias, small sample size, and lack of text utilization in the model. Addressing these challenges could involve:

1. **Temporal ROI and Labels:** CAM results show the model focuses on articulation errors, though some outputs are unclear. Incorporating temporal labels can guide the model to relevant decision regions, akin to adding segmentation in medical image tasks.
2. **Leveraging Large Language Models (LLMs):** LLMs like QWEN2.5 or OpenAI's Whisper-large-v2 can better link transcriptions to disease states. Adapting these models requires:
 - **Metric Alignment:** Adjust log-loss metrics for binary logits instead of token predictions.
 - **Text Similarity:** Collect transcriptionally similar samples to align with LLM strengths.

Implementing these changes can improve model accuracy, interpretability, and generalizability to clinical scenarios.

Moreover, looking forward, integrating this model with wearable devices could enable real-time monitoring of speech patterns, providing early warnings for cognitive decline. This would represent a novel, scalable approach to addressing AD/ADRD in diverse populations.

6. References

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